

Machine Learning Based Buyer Segmentation and Investment Profiling

for Real Estate Market Intelligence

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In collaboration with Parcel Co. Limited

Interactive Dashboard: <https://parcel-buyer-segmentation-ml-6xrw5zwx9tsscld4lsgu6c.streamlit.app>

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Abstract

This research paper presents a machine learning-driven approach to buyer segmentation and investment profiling for real estate market intelligence. The study utilizes K-Means clustering and hierarchical clustering algorithms to identify distinct buyer segments within the Parcel real estate platform. By analyzing demographic attributes, acquisition patterns, geographic distribution, and financial behaviors of 2,000 property buyers, we identified four distinct buyer segments: Global Investors, First-Time Buyers, Corporate Buyers, and Luxury Investors. The segmentation model achieved a silhouette score of 0.120, demonstrating meaningful separation between clusters. Our findings reveal significant differences in spending patterns, investment motivations, and financing preferences across segments, providing actionable insights for targeted marketing strategies and personalized property recommendations. The results demonstrate how AI-driven buyer intelligence can transform traditional real estate analytics by uncovering hidden customer segments that conventional approaches fail to detect.

Keywords: Buyer Segmentation, K-Means Clustering, Real Estate Analytics, Investment Profiling, Machine Learning, Market Intelligence

1. Introduction

The real estate industry operates in a highly heterogeneous market where buyer behavior varies significantly across demographics, geographies, and financial profiles. Understanding these distinct buyer types is crucial for developing effective marketing strategies, personalized property recommendations, and data-driven investment decisions.

Real estate companies like Parcel interact with diverse buyer categories including individual home buyers, institutional investors, international buyers, high-net-worth individuals, and first-time buyers. Without proper segmentation, companies treat all buyers uniformly, leading to inefficient marketing spend, poor customer targeting, and missed investment opportunities.

This research addresses the critical need for data-driven buyer intelligence by applying machine learning clustering techniques to identify natural groupings of property buyers. The primary objectives of this study are to:

- Identify distinct buyer segments based on demographic, financial, and behavioral attributes

- Characterize each segment's investment patterns, geographic preferences, and financing behaviors
- Develop actionable profiles that enable targeted marketing and personalized recommendations
- Validate the clustering approach using multiple evaluation metrics and hierarchical validation

2. Literature Review

Customer segmentation has been a cornerstone of marketing strategy for decades. Traditional approaches rely on demographic variables such as age, income, and geographic location to classify customers into meaningful groups. However, the advent of machine learning has enabled more sophisticated segmentation approaches that can discover patterns invisible to conventional methods.

2.1 Clustering in Market Segmentation

Clustering algorithms have been widely adopted in market segmentation across various industries. K-Means clustering, first introduced by Lloyd (1957) and later refined by MacQueen (1967), remains one of the most popular unsupervised learning techniques due to its computational efficiency and interpretability. The algorithm partitions data into K clusters by minimizing the within-cluster sum of squares (WCSS).

Hierarchical clustering provides an alternative approach that builds a tree-like structure (dendrogram) representing nested clusters. Ward's method, which minimizes the variance within clusters, has proven particularly effective for market segmentation tasks where compact, well-separated clusters are desired.

2.2 Real Estate Analytics

The application of machine learning in real estate has grown significantly in recent years. Studies have demonstrated the value of predictive modeling for property valuation, demand forecasting, and buyer behavior analysis. However, buyer segmentation in real estate remains relatively underexplored compared to other domains such as retail and finance.

Research by Park and Bae (2015) highlighted the importance of integrating multiple data sources for effective customer segmentation. In real estate, this includes not only

demographic data but also transaction history, property preferences, financing patterns, and geographic mobility.

2.3 Investment Profiling

Understanding investor behavior is critical for real estate platforms that serve both end-users and investment buyers. Investment profiling involves characterizing buyers based on their risk tolerance, investment horizon, portfolio diversification, and return expectations. Machine learning techniques enable the automated discovery of investor profiles from historical transaction data.

3. Methodology

3.1 Data Collection and Preprocessing

The dataset comprises 2,000 client records from the Parcel real estate platform, containing demographic attributes, transaction history, and behavioral indicators. The primary data sources include client registration data and property transaction records.

Data preprocessing involved several critical steps. Duplicate records were identified and removed. Date fields were standardized and age was calculated relative to a reference date of January 1, 2024. Categorical variables including client type, gender, country, region, acquisition purpose, loan application status, and referral channel were normalized through title-casing and stripping of extraneous whitespace.

The sale price field, originally stored as formatted currency strings, was converted to numeric values by removing dollar signs and commas. Transaction dates were parsed into consistent datetime formats. Only property listings with 'Sold' status and valid client references were retained for analysis, resulting in a clean dataset of 2,000 buyers with complete transaction histories.

3.2 Feature Engineering

A comprehensive set of features was engineered to capture the multidimensional nature of buyer behavior. Binary encoding was applied to categorical variables: loan application status (Yes/No), acquisition purpose (Investment/Home), client type (Company/Individual), and gender (Male/Female).

Multi-category variables including country, referral channel, and region were encoded using Label Encoding. For clustering, the final feature set comprised 11 variables: age,

satisfaction score, loan applied, investment indicator, corporate indicator, gender, country, referral channel, total spend, units bought, and average area in square feet.

All features were standardized using `StandardScaler` to ensure equal contribution to the distance calculations in the clustering algorithm. This transformation ensures each feature has approximately zero mean and unit variance, preventing variables with larger scales from dominating the clustering process.

3.3 Clustering Algorithms

Two complementary clustering approaches were employed in this study. K-Means clustering was used as the primary segmentation algorithm due to its scalability and interpretability. The algorithm was configured with 10 random initializations (`n_init=10`) and a fixed random seed (`random_state=42`) to ensure reproducibility.

Hierarchical clustering with Ward's linkage method was applied as a validation technique. A sample of 300 observations was used for dendrogram visualization to ensure readability while preserving the overall cluster structure. The dendrogram provides visual confirmation of the natural cluster hierarchy and validates the K-Means results.

The optimal number of clusters was determined using a combination of the Elbow Method and Silhouette Analysis. The Elbow Method examines the rate of decrease in within-cluster sum of squares (WCSS) as the number of clusters increases, identifying the point where additional clusters provide diminishing returns. The Silhouette Score measures how similar an object is to its own cluster compared to other clusters, with values ranging from -1 to +1.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distributions, relationships, and patterns within the dataset before applying clustering algorithms. This phase provides critical insights that inform feature selection and clustering parameter choices.

4.1 Buyer Attribute Distributions

The analysis of buyer attribute distributions reveals important characteristics of the dataset. Age distribution spans from approximately 20 to 90 years, with a relatively uniform distribution across the middle-age range. This diversity in age groups suggests the presence of distinct generational segments within the buyer population.

Satisfaction scores are distributed across all five rating levels (1-5), with a slight concentration in the middle range. Client type analysis shows that individual buyers constitute 94.8% of the dataset, while corporate buyers represent 5.2%. This skew toward individual buyers is typical for residential real estate platforms.

Acquisition purpose reveals that 70.2% of buyers purchased for home use, while 29.8% bought for investment purposes. Loan application patterns show that 36.8% of buyers applied for financing, while 63.2% did not require loans. The referral channel analysis indicates that Website is the dominant acquisition channel, followed by Agency and Client referrals.

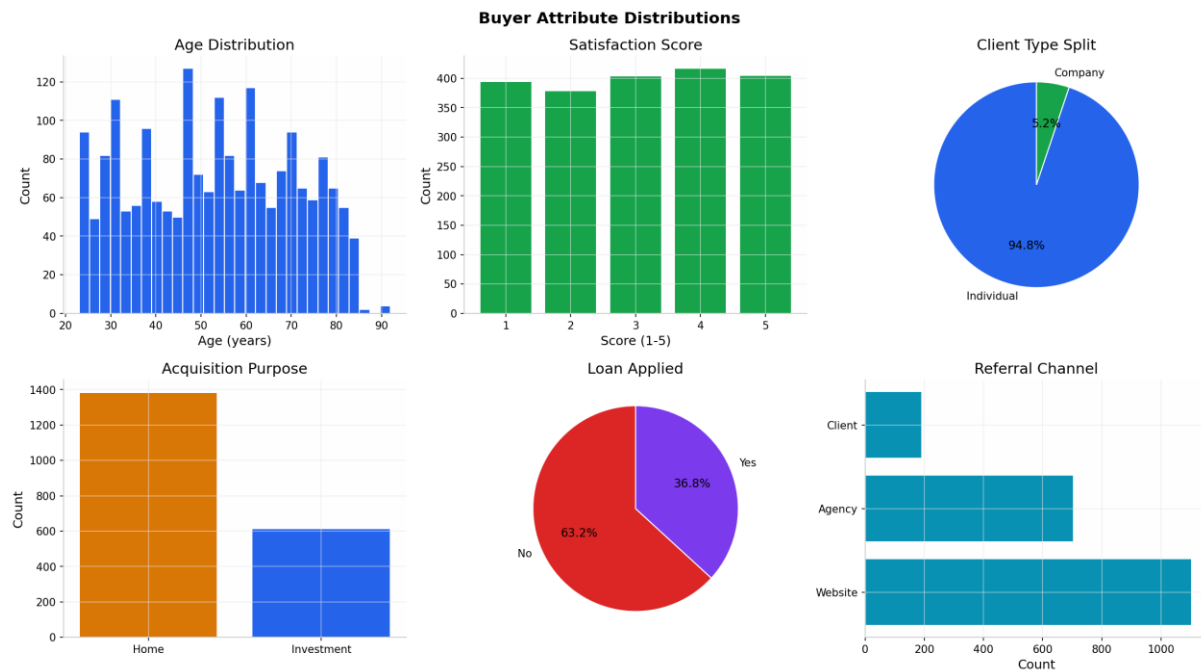


Figure 1: Buyer Attribute Distributions - Comprehensive overview of demographic and behavioral attributes across the buyer population

4.2 Geographic Analysis

Geographic analysis reveals the international scope of the Parcl platform. The United States dominates with approximately 75% of all buyers, followed by the United Kingdom, Canada, Germany, and France. Additional countries include Belgium, Mexico, Australia, Russia, and Denmark, demonstrating the platform's global reach.

Within the United States, California leads by a significant margin with over 600 buyers, followed by Nevada, Colorado, Arizona, Oregon, and Utah. This concentration in western states reflects both the platform's market penetration and the appeal of real estate investment in these regions.

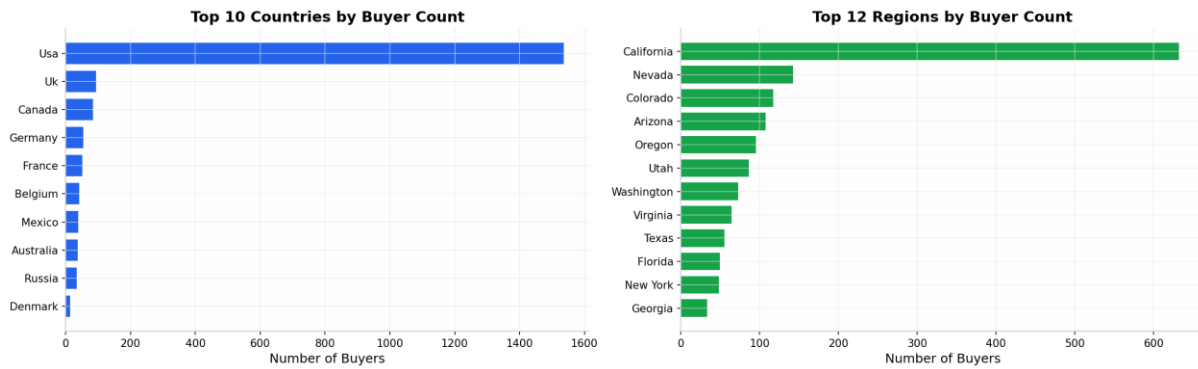


Figure 2: Geographic Distribution - Top 10 countries and 12 regions by buyer count, showing the platform's global reach

4.3 Investment vs Home Analysis

The comparison between investment and home buyers reveals distinct spending patterns. Boxplot analysis shows that both groups have similar median spending levels around \$1.2 million, but investment buyers exhibit greater variance with several high-value outliers exceeding \$3.5 million. This suggests a subset of investment buyers making significantly larger purchases.

Loan application patterns differ notably between acquisition purposes. Home buyers show a higher rate of loan applications (approximately 36%) compared to investment buyers (approximately 22%). This difference reflects the distinct financing strategies employed by residential buyers versus investors, who may leverage alternative funding sources.

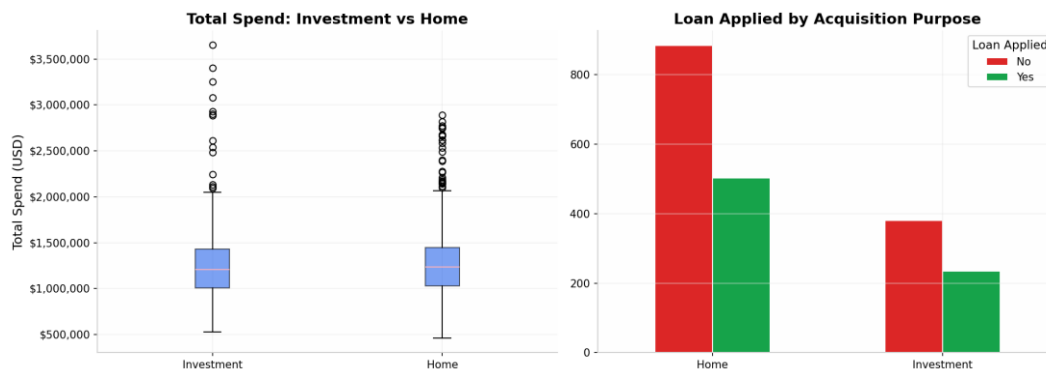


Figure 3: Investment vs Home Analysis - Spending comparison and loan application patterns by acquisition purpose

4.4 Correlation Analysis

The correlation heatmap reveals several meaningful relationships among numeric features. The strongest correlation exists between total spend and units bought ($r = 0.71$), which is expected as buyers purchasing more units naturally incur higher total expenditures.

Average area shows a moderate positive correlation with total spend ($r = 0.57$), indicating that buyers of larger properties tend to spend more.

Age shows minimal correlation with most financial variables, suggesting that spending behavior is not strongly age-dependent. Satisfaction score also shows weak correlations with other variables, indicating that customer satisfaction is influenced by factors beyond the numeric features captured in this dataset.

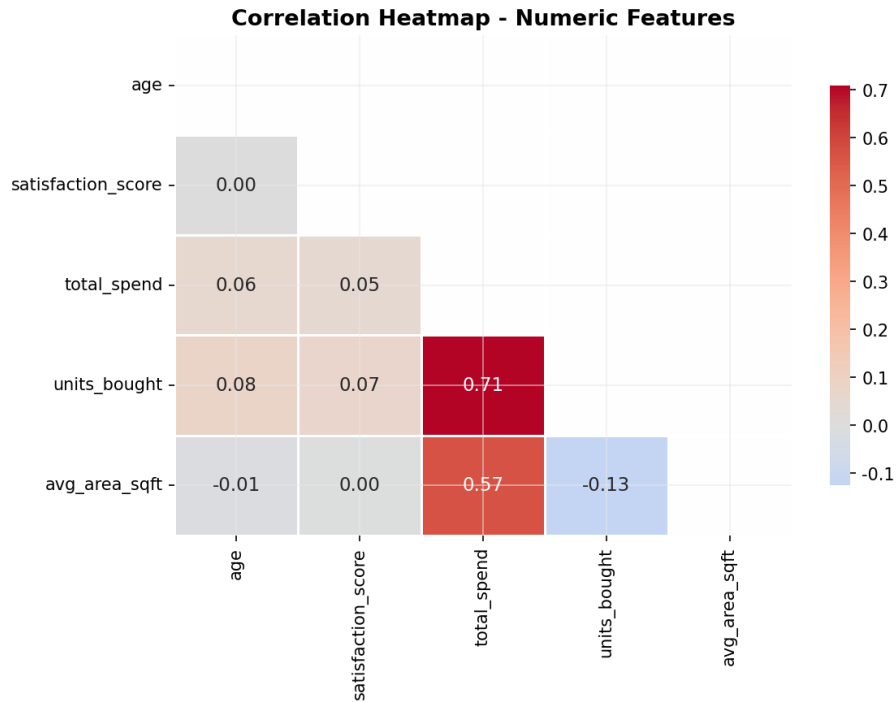


Figure 4: Correlation Heatmap - Relationships among numeric features reveal key patterns in buyer behavior

5. Clustering Analysis and Results

This section presents the core clustering analysis, including optimal cluster selection, algorithm validation, segment visualization, and detailed profile characterization.

5.1 Optimal Cluster Selection

The Elbow Method and Silhouette Score analysis were employed to determine the optimal number of clusters. The Elbow plot shows a noticeable bend at $K=4$, where the rate of WCSS reduction begins to diminish. Beyond $K=4$, additional clusters provide only marginal improvements in within-cluster compactness.

The Silhouette Score analysis corroborates this finding, showing a local peak at $K=4$ with a score of 0.1199. While $K=6$ achieves a slightly higher silhouette score, the difference

is minimal and $K=4$ provides better interpretability and business relevance. Based on these complementary evaluation methods, $K=4$ was selected as the optimal number of clusters.

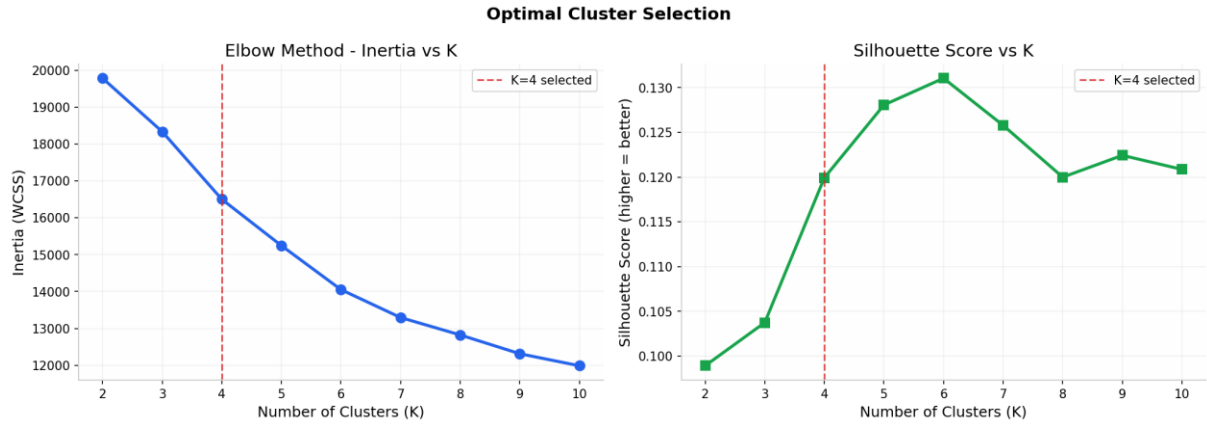


Figure 5: Optimal Cluster Selection - Elbow Method and Silhouette Score analysis confirming $K=4$ as the optimal number of clusters

5.2 Hierarchical Clustering Validation

The hierarchical clustering dendrogram provides visual validation of the four-cluster solution. Using Ward's linkage method on a representative sample of 300 observations, the dendrogram clearly shows four major branches that merge at a distance threshold of approximately 20.5. This structure confirms the natural existence of four distinct buyer groups within the data.

The dendrogram also reveals the relative similarity between clusters. The two closest clusters merge at a relatively low distance, suggesting some overlap in their characteristics, while the most distant clusters remain separated until much higher merge distances, indicating well-differentiated segments.

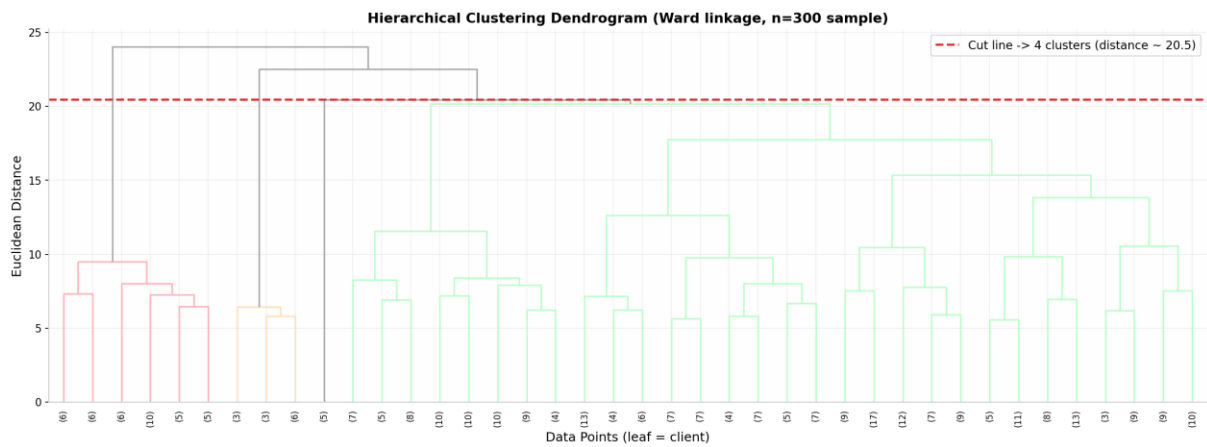


Figure 6: Hierarchical Clustering Dendrogram - Ward linkage visualization confirming four natural clusters in the buyer population

5.3 PCA Visualization

Principal Component Analysis (PCA) was applied to project the 11-dimensional feature space into two dimensions for visualization. The first two principal components explain 27.8% of the total variance, which is sufficient to visualize the cluster structure while acknowledging that much of the variance exists in higher dimensions.

The PCA scatter plot reveals clear separation between the four segments. C4 (Luxury Investors) occupies the right portion of the plot, distinguished primarily by higher total spend and greater unit purchases. C1 (Global Investors) spans the left-central region, showing diversity in geographic origins. C2 (First-Time Buyers) clusters in the lower-central area, characterized by younger age and loan dependency. C3 (Corporate Buyers) appears as a smaller, distinct cluster reflecting their unique organizational buying patterns.

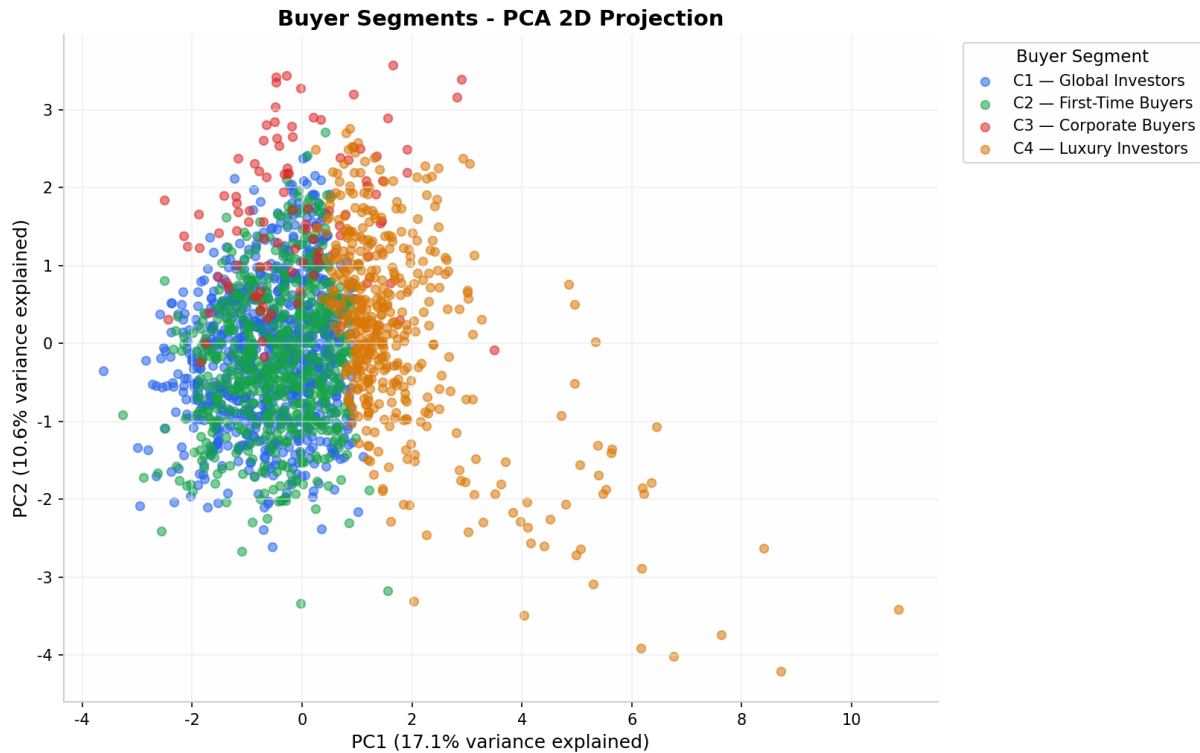


Figure 7: PCA 2D Projection - Four buyer segments visualized in principal component space, showing distinct cluster separation

5.4 Cluster Profiles

Detailed cluster profiles were constructed by analyzing mean values of key features for each segment. The profiles reveal distinct characteristics that enable clear differentiation between buyer types.

C1 - Global Investors represent international buyers with investment goals. They exhibit moderate spending (~\$1.1M), diverse geographic origins, and a balanced mix of home and investment purchases. This segment shows the highest diversity in country representation.

C2 - First-Time Buyers are younger buyers (average age 53.2) relying on loans for personal home purchases. They show moderate satisfaction scores and primarily acquire properties through website referrals. Their spending is moderate at approximately \$1.1M.

C3 - Corporate Buyers are companies purchasing multiple units including office spaces. They represent the smallest segment (5.2% of buyers) but show distinct behavioral patterns with high loan application rates and investment-focused acquisitions.

C4 - Luxury Investors are high-spend investors (average \$1.7M) with top satisfaction scores and premium property focus. They purchase the most units on average (4.3) and show the highest spending per transaction.

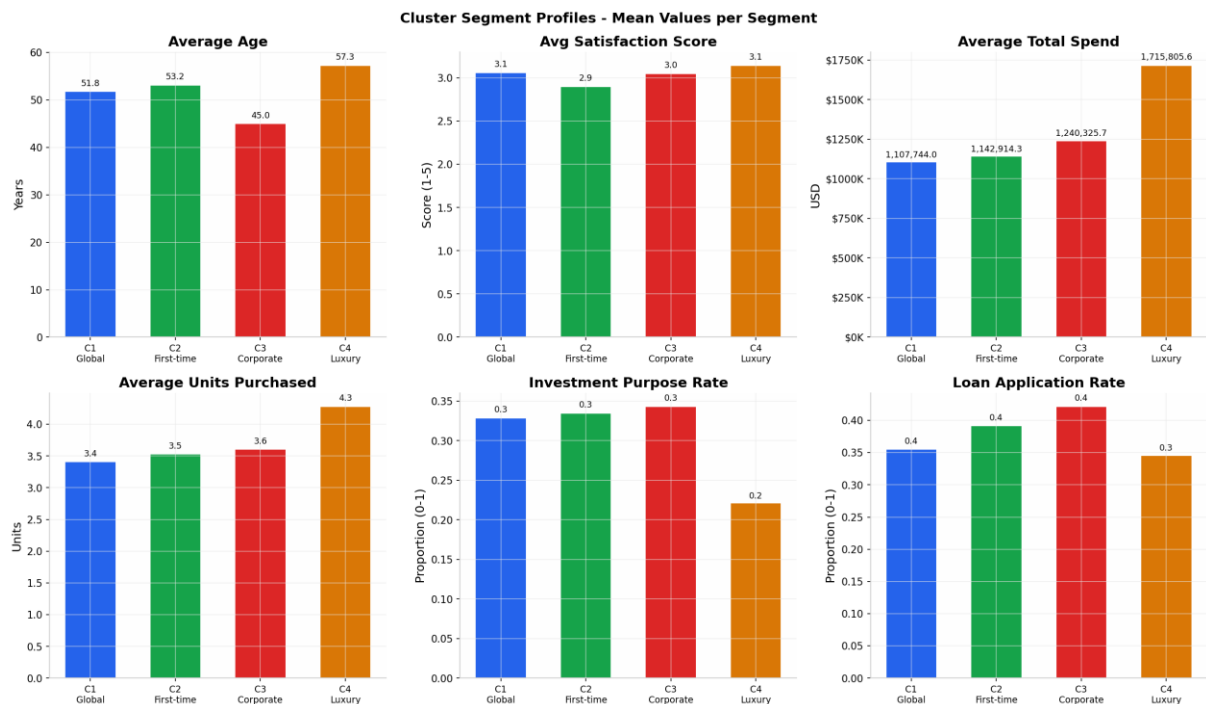


Figure 8: Cluster Segment Profiles - Mean values of key attributes across the four identified buyer segments

5.5 Segment Behavior Analysis

Behavioral analysis reveals distinct acquisition patterns across segments. Global Investors show a relatively balanced split between home and investment purchases, reflecting their diversified portfolio approach. First-Time Buyers predominantly purchase homes, consistent with their buyer profile. Corporate Buyers are almost entirely investment-focused,

while Luxury Investors show a higher proportion of home purchases with premium spending levels.

Referral channel preferences also vary significantly. Website referrals dominate across all segments, but the distribution varies. First-Time Buyers show the strongest reliance on website channels, while Corporate Buyers have a more balanced distribution across Agency and Website channels.

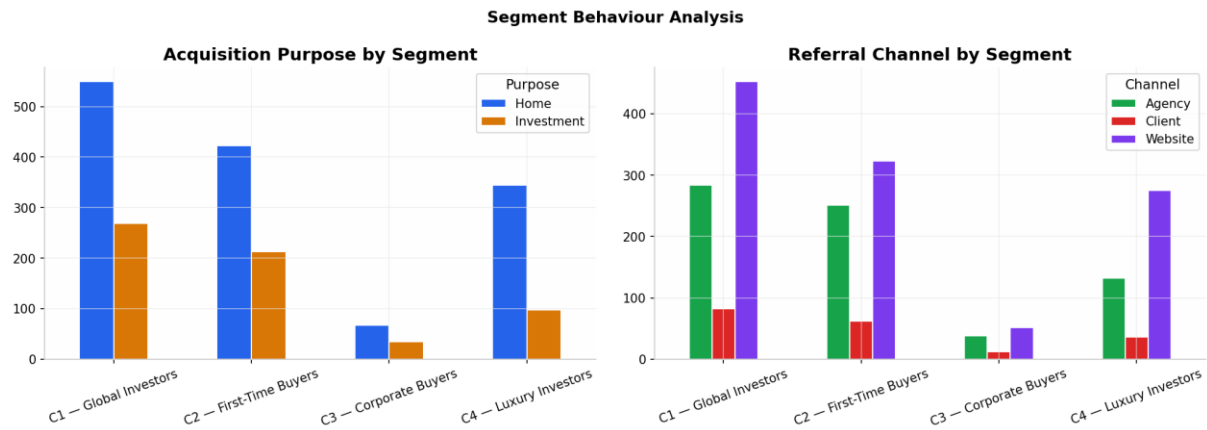


Figure 9: Segment Behavior Analysis - Acquisition purpose and referral channel distribution by buyer segment

5.6 Silhouette Analysis

The silhouette analysis provides a detailed quality assessment of the clustering solution. The average silhouette score of 0.120 indicates that clusters are moderately well-separated, with most observations showing positive silhouette coefficients.

Individual cluster analysis reveals that C1 (Global Investors) and C3 (Corporate Buyers) achieve the highest silhouette scores, indicating strong internal cohesion and clear separation from other segments. C2 (First-Time Buyers) shows more overlap with adjacent clusters, reflecting the natural continuum between first-time and experienced buyers. All clusters maintain predominantly positive silhouette values, confirming the validity of the four-segment solution.

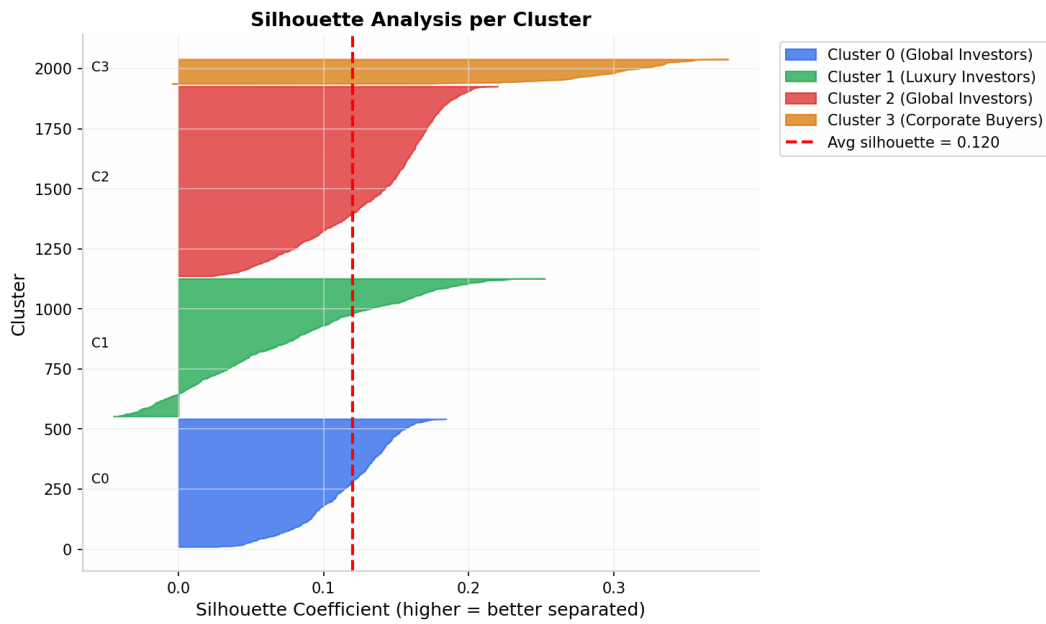


Figure 10: Silhouette Analysis per Cluster - Quality assessment showing positive separation across all four buyer segments

6. Discussion

The clustering analysis successfully identified four distinct buyer segments that align with real-world real estate market dynamics. Each segment exhibits unique characteristics that have direct implications for marketing strategy, customer engagement, and platform development.

The Luxury Investors segment (C4) represents the highest-value customers with average spending of \$1.7 million. Their premium property focus and high satisfaction scores suggest that targeted luxury offerings and personalized concierge services could further enhance their experience and lifetime value.

Global Investors (C1) demonstrate the importance of international market expansion. Their diverse geographic origins indicate opportunities for region-specific marketing campaigns and multilingual platform features. Cross-border investment tools and international financing partnerships could strengthen engagement with this segment.

First-Time Buyers (C2) represent a critical growth segment. Their loan dependency and moderate spending suggest the need for financial advisory services, mortgage partnerships, and entry-level property showcases. Converting these buyers into repeat customers could significantly impact long-term revenue.

Corporate Buyers (C3), while the smallest segment, offer unique opportunities for B2B relationship development. Their multi-unit purchases and investment focus suggest potential for bulk transaction services, portfolio management tools, and commercial property listings.

7. Conclusion and Recommendations

This research demonstrates the effectiveness of machine learning clustering for buyer segmentation in real estate. The four-segment model provides a robust framework for understanding customer diversity and tailoring business strategies accordingly.

The study makes several contributions to the field. First, it validates the application of K-Means clustering combined with hierarchical validation for real estate buyer segmentation. Second, it provides detailed segment profiles that can be directly operationalized by real estate platforms. Third, it demonstrates how PCA visualization can effectively communicate cluster characteristics to business stakeholders.

Based on the findings, the following recommendations are proposed:

- Develop segment-specific marketing campaigns tailored to the investment motivations and communication preferences of each buyer group
- Implement personalized property recommendation engines that account for segment-specific preferences in location, price range, and property type
- Create dedicated service tiers for Luxury Investors and Corporate Buyers with specialized account management
- Establish financial partnerships to support First-Time Buyers with mortgage pre-approval and financing guidance
- Expand international marketing efforts targeting Global Investors with region-specific content and local market insights

Future research directions include incorporating temporal dynamics to track segment evolution over time, integrating additional data sources such as property browsing behavior and social media engagement, and developing predictive models to identify segment membership for new customers at registration.

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